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Efficient catalytic conversion of ethanol to 1-butanol via the Guerbet reaction over copper- and nickel-doped porous metal oxides

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Abstract

The direct conversion of ethanol to higher value 1-butanol is a catalytic transformation of great interest in light of the expected wide availability of bioethanol originating from the fermentation of renewable resources. In this contribution we describe several novel compositions of porous metal oxides (PMO) as highly active and selective catalysts for the Guerbet-coupling of ethanol to 1-butanol in the temperature range of 180-320 °C. The novel PMO catalysts that do not contain any noble metals are obtained by calcination of a series of hydrotalcite precursors synthesized through modular procedures. In particular, catalyst compositions simultaneously containing Cu and Ni dopants have shown excellent catalytic activities. Up to 22% 1-butanol yield at 56% ethanol conversion were reached in a batch mode, recycling and leaching tests showed excellent robustness of the new catalysts. An extensive characterization by means of several techniques such as powder XRD, SEM, TEM, BET, NH_3 - and CO_2 -TPD was performed in order to understand structure-activity trends.

Key words:

Bio-ethanol, 1-butanol, Guerbet reaction, copper, porous metal oxides

Introduction

Ethanol obtained by fermentation of renewable biomass currently represents one of the largest volume liquid biofuels produced worldwide.^{1,2} Owing to the recently established biorefineries capable of processing non-edible lignocellulose feedstocks such as agricultural or forestry waste, its amount is expected to increase by 4-6% annually to 30 billion gallons in the next 5 years.³ The most common application of bioethanol is its use as blending agent with gasoline; however a relatively low energy density compared to gasoline (24 MJ/L vs 34.2 MJ/L) and miscibility with water restrict its application in the transportation sector.⁴ As an excellent alternative, 1-butanol possesses an energy density closer to gasoline (29.2 MJ/L), and poor solubility in water (74g/L at 25 °C)⁵. Moreover, 1-butanol can also be converted to C4-olefins (butenes), which are precursors to other valuable commodity chemicals⁶. Therefore the efficient conversion of ethanol to 1-butanol is of great interest for the modern bio-based economy. A direct chemical route to 1-butanol relies on the self-coupling of ethanol through the Guerbet reaction⁷. Innovative homogeneous catalytic methods using Ru⁸⁻¹⁰ and Ir¹¹ complexes have shown excellent selectivity towards 1-butanol under relatively mild reaction conditions.

Heterogeneous catalysts reported for this transformation have been recently summarized in excellent reviews.^{7,12} An overview of selected results in the Guerbet coupling of ethanol to 1-butanol is also summarized in Table S1 and Table S2 with the corresponding references.

Frequently used heterogeneous catalysts reported for the Guerbet reaction include MgO¹³, Mg/Al mixed metal oxides¹⁴ and hydroxyapatites^{15,16} that were applied in both continuous and batch processes (see also Table S1 and further references). Several mechanistic studies involving these classes of catalysts pointed towards the importance of surface properties, especially acidity and basicity as these influence several key reaction steps, including ethanol dehydrogenation.¹²⁻¹⁴ The right balance of acidic and basic sites influences catalytic activity and also product selectivity, since the product 1-butanol can undergo several side reactions, mainly dehydration^{16,17}. Accordingly, a dependence of product selectivity on Mg/Al ratio in hydrotalcite-derived Mg-Al mixed oxides has

been observed.^{14,18} In addition to varying the Mg to Al ratio, the composition of Mg-Al mixed oxides can be easily modified by introducing additional metal dopants in order to tune their catalytic activity. Although there are some literature examples of doped Mg-Al mixed oxides in the vapor¹⁹ and liquid phase²⁰, guerbet-reaction of ethanol, a highly efficient system has not yet been devised.

In this contribution we describe several novel PMOs derived from synthetic hydrotalcites that do not contain any noble metals. Replacing a small portion of Mg^{2+} in the catalyst structure with Cu- and Ni- resulted in catalyst compositions that are highly efficient in the Guerbet-coupling of ethanol to 1-butanol. The catalysts are robust over several cycles and have been extensively characterized.

Experimental procedure

Catalyst Preparation

The HTC (hydrotalcite) catalyst precursors were prepared by a coprecipitation method, according to literature procedures^{21,22}. In a typical procedure, a solution containing $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ (0.06mol, 12.2g), $\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$ (0.025mol, 6.0g), $\text{Cu}(\text{NO}_3)_2 \cdot 2.5\text{H}_2\text{O}$ (0.0075mol, 1.74g), $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (0.0075mol, 2.18g) in deionized water (0.1 L) was slowly added to an aqueous solution (0.15 L) of Na_2CO_3 (0.025mol, 1g) at 60 °C under vigorous stirring. The pH was carefully maintained between 9 and 10 by adjusting with frequent additions of an aqueous solution of NaOH (1 M). The mixture was vigorously stirred for 72h at 60°C. After cooling to room temperature, the suspension was filtered, and the solid was washed with deionized water and resuspended into a solution of Na_2CO_3 (2M) which was stirred for 24 hours at 40°C. After the catalyst precursor was filtered and washed with deionized water until the washings were chloride-free. The solid was dried at 120 °C overnight, and the hydrotalcite precursor (HTC) was obtained as blue powder (6.68 g). The corresponding copper-nickel-doped porous metal oxide (Cu10Ni10-PMO) was obtained after calcining the HTC material at 460 °C for 24 h in air. All porous metal oxides used in this paper were prepared according to the same procedure, replacing a defined amount of Mg^{2+} with Ni^{2+} and/or Cu^{2+} as specified below.

Catalysts characterization

Powder X-ray analysis was performed on a Bruker XRD diffractometer using Cu K α radiation and the spectra was recorded in the 2 θ angle range of 5°-70°. Elemental analyses were performed on a Perkin Elmer instrument (Optima 7000DV) after full solubilization of the PMO catalysts in *aqua regia*. The textural characterization was achieved using conventional nitrogen adsorption/desorption method, with a Micromeritics ASAP 2420 automatic analyze. Prior to nitrogen adsorption, the samples were outgassed for 8 h at 250 °C. Surface concentration of acidic and basic sites was determined by using a linear quartz micro-reactor (*l*, 200 mm; *i.d.*, 4 mm) in a conventional flow apparatus operating both in continuous and pulse mode. TEM measurements were performed on a Philips CM12 instrument equipped with a high-resolution camera. Powdered samples were dispersed in 2-propanol under ultrasound irradiation and the resulting suspension put drop-wise on a holey carbon-coated support grid. Elemental distribution was measured in STEM mode on another machine (FEI Tecnai T20 electron microscope) using a HAADF detector and a X-max 80 EDX detector (Oxford instruments). The morphology of the samples was investigated by scanning electron microscopy measurements carried out by using a Philips XL-30-FEG Scanning Electron Microscope at an accelerating voltage of 5-15 kV. Specimens were deposited as powders on aluminum pin flat stubs.

Reaction

In a typical experiment, the catalyst (0.1 g unless otherwise stated) was placed in a Swagelok stainless steel microreactor (10 mL) and ethanol (3 mL) and decane (20 μ L, internal standard) were added. The reactor was sealed and placed in an aluminum block pre-heated at the desired temperature (NOTE: Autogeneous pressure of cca 80 bar is generated). After the indicated reaction time, the microreactor was cooled down with an ice-water bath and subsequently carefully opened (CAUTION: release of pressurized gases! For more details on the gas phase see Supporting Information, page S3). The liquid sample as well as the catalyst was quantitatively transferred to a 50

mL centrifuge tube. Additional THF was used to wash the reactor and recover all catalyst residues (up to 10 mL total volume). After centrifugation, the solution was transferred into a glass vial and analyzed by GC-FID (Hewlett Packard 6890 series equipped with a HP-5 capillary column) in order to determine the ethanol conversion and product yields. Compounds were also identified by GC-MS and the injection of pure reference standards for the comparison of retention times in the GC column.

Results and discussion

Synthesis and Characterization of porous metal oxides

The hydrogen neutral reaction sequence of the Guerbet-coupling of ethanol to 1-butanol involves the following key steps: a) dehydrogenation of ethanol to acetaldehyde, b) aldol condensation of acetaldehyde including dehydration to afford the corresponding unsaturated C4 products and c) hydrogenation to form saturated longer chain alcohols (Scheme 1)^{7,12}. An ideal catalyst should thus possess hydrogenation/dehydrogenation activity in addition to facilitating the aldol-addition and dehydration steps. Previous reports from the literature confirmed the favorable behavior of Mg/Al-PMO in promoting the self-coupling of ethanol.¹⁴ Based on our previous experiences with Cu-doped PMO in a variety of transformations related to hydrogenation and dehydrogenation of biomass derived alcohols and polyols^{21,23}, we expected that such catalyst compositions might facilitate the first and last step of the Guerbet reaction cycle and thus be superior to the common Mg/Al-hydrotalcite structure. Especially a facile dehydrogenation step would be beneficial that has been previously proposed to be rate determining⁷. Therefore we set to synthesize a variety of different catalyst compositions to be tested in the Guerbet coupling of ethanol to 1-butanol based on modification of the basic Mg/Al hydrotalcite structure.

First, hydrotalcites (HTC) were synthesized, all containing small amount of Cu and/or Ni dopant in place of a portion of Mg²⁺ while the M²⁺/M³⁺ ratio was kept at a constant 3:1. The formation of the double-layered structure during HTC synthesis was confirmed by powder XRD measurements (Figure

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3 1, left). These exhibited sharp diffraction peaks at 11.4°, 23.21°, 34.73° and broad reflections at
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5 39.20°, 46.50°, 52.44°, 60.61° and 61.81° ascribed to the <003>, <006>, <012>, <015>, <018>, <110>
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7 and <113> planes referred to the trigonal hydrotalcite [JCPDS, 350965] according to reported data.²⁴
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10 The synthetic HTC were subsequently calcined at 460°C overnight, during which the lamellar
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12 hydrotalcite structure was converted into an amorphous mixed-oxide composition, where a residual
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14 Mg(Al)O phase is visible (peaks at 43) whereas no CuO (peaks at 23, 28 and 31) nor NiO oxide phase
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16 (peaks at 38, 44 and 64) could be detected (Figure 1, right), indicating a homogeneous dispersion of
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18 both active metals. Their composition, determined by ICP analysis, was found in very good
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20 agreement with the theoretically expected values indicating good incorporation of the metal ions
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22 into the parent hydrotalcite structures (Table 1 and Table S3).
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26 The specific surface area of all catalysts, as determined by BET measurements, was comparable
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28 within small variations. As shown in Table 1, the addition of Cu slightly decreased the BET to 197.7
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30 m²/g according to a trend observed by other groups²⁵, while the simultaneous presence of nickel and
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32 copper (255.7 m²/g) or nickel alone (239.2 m²/g) did not change significantly the total surface area
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34 compared to the reference MgAl-PMO (233.8 m²/g). Also the pore volume increased for Cu₁₀Ni₁₀-
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36 PMO (1.06 cm³/g) compared to the MgAl-PMO catalyst (0.91 cm³/g).
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41 ***Reactivity of PMO catalysts in the Guerbet reaction***

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43 We have started our investigations with comparing the reactivity of the various PMO catalysts at 310
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45 °C for 6 hours in a stainless steel batch minireactor. As expected, marked differences in reactivity
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47 were seen depending on the nature of the dopant. The results obtained are summarized in Figure 2
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49 and Table 2. With MgAl-PMO, not containing any additional dopants, a 5.6 % 1-butanol yield was
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51 observed at 30.8% ethanol conversion in 6 hours, alongside with other higher alcohols such as 2-
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53 ethyl-hexanol, 1-hexanol, 1-octanol (Table 2, Entry 1). The other large group of products was esters.
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55 These esters may be formed by the dehydrogenation of a possible hemiacetal intermediate that is
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57 likely formed under the reaction conditions (see Scheme 1). All these by products can in principle
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59 also be used as fuels²⁶ or as blending agents in gasoline²⁷. Besides alcohols and esters, diethyl ether
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as well as acetaldehyde, which formed by dehydration and dehydrogenation reactions, was also detected by GC-MS (see Table S4). Ethylene was detected in the gas phase (Supporting Information, page S3). Based on previous research²⁸, it is known that copper as a dopant could significantly speed up the rate of ethanol dehydrogenation, which is the first step towards the formation of 1-butanol. Indeed, with Cu₂₀-PMO (Table 2, Entry 2) ethanol conversion was increased to 37.8%, however the 1-butanol yield (5.7%) was practically identical to that observed with the MgAl-PMO (Table 2, Entry 2 vs 1). This is due to the formation of ethyl-acetate as main product (18.1% yield), which would correspond to a favored dehydrogenation of a possible hemiacetal intermediate²⁹. The use of nickel only as dopant slightly increased the 1-butanol yield to 8.8% (Table 2, Entry 3) but other long chain alcohols remained the main products. The best catalytic performance was achieved with catalysts containing both copper and nickel. The conversion values were increased to 62.4%, 51.2% and 47.9% (Table 2, Entry 4, 5 and 6) and 1-butanol yield was 18.7%, 17.9%, and 21.1%, respectively (Table 2, Entry 4, 5 and 6). With the increasing amount of Ni dopant besides copper, the selectivity of 1-butanol remained at a constant value of 72%, but the total alcohol to ester ratio increased. The space-time yields with these Cu-Ni PMOs were approximately 3 times higher than those obtained with the composition with no dopants. A control reaction performed under argon atmosphere showed no difference (Table 2, Entry 7). The HTC alone was also active, but the yield of 1-butanol was much lower (Table 2, Entry 8). To our knowledge, PMO catalysts containing both Ni and Cu dopants have not yet been used for the Guerbet coupling of ethanol. A few examples of similar materials include the hydrogenation of glucose to sorbitol³⁰, the transesterification of dimethyl oxalate with phenol³¹, steam reforming reactions^{32,33} and furfural hydrogenation³⁴. None of these reports have addressed the influence of the mutual amounts of both dopants in the catalytic outcome. In our study it appears that the ratio between Cu and Ni has an influence on the catalytic performance and the catalyst composition with equimolar amounts of Cu and Ni lead to the best, 21.1% yield and 72% selectivity of 1-butanol (Table 2, Entry 6) among the tested catalysts.

Next, this catalyst composition was studied in greater detail. On Figure 3 and Table S5 the dependence of the conversion and yield values on the reaction temperature (in the range 180 - 320 °C) and catalyst loading (ranging from 0.05 g to 0.2 g) is shown. As expected, the conversion consistently increased from 2.3 % at 180 °C to 56.5 % at 320 °C. At the same time, the yield of 1-butanol increased up to a maximum of 22.2 % and space time yield reached the outstanding value of 704.6 g kg_{cat}⁻¹h⁻¹ at 320 °C. However, the carbon balance decreased from 98.5% to 75.0% with the increase of temperature (Table S3). This is attributed to the formation of volatile by-products (See also Supporting Information, Page S3). It should be noted that the runs at 180 °C and 220 °C take place in liquid phase, while the reactions at higher temperature are under supercritical conditions, which should facilitate mass transfer. The yield of long chain alcohols rose with the temperature accordingly (Figure 3), whereby the formation of C6+ products (especially C6 and C8 alcohols) at higher temperatures was mainly due to the self-coupling of 1-butanol and to the coupling of 1-butanol with ethanol. That is why the selectivity of 1-butanol decreased from 78% (Table S5, Entry 2 vs 6) to 70% but the ratio of alcohols to esters rose from 5.9 to 14.1. With increasing catalyst loading, conversion values (62.2%) have improved but no major changes in the 1-butanol yield took place due to the competing formation of C6 and C8 alcohols (Figure 3).

Next, product formation profiles for 24 hours were recorded at 310 °C with 0.1 g of Cu10Ni10-PMO catalyst. As shown in Figure 4, the conversion of ethanol displayed a linear increase up to approximately 10 hours, after which no meaningful difference was observed. The yield of 1-butanol reached a constant value of 21% after 6h. The increase in ethanol conversion in the first 10 hours corresponded to the competing conversion of 1-butanol to higher alcohols, especially between 6 and 10 hours.

Catalyst recycling

Recycling experiments using the Cu10Ni10-PMO found good robustness even after seven cycles (Figure 5 and Table S6). The catalyst was recovered at the end of each run, washed with THF and acetone, and reused after drying. A variation in the 1-butanol yield was visible between the first and

the second cycle (21% vs. 14%) while the relative product distribution remained constant for the subsequent cycles, and 1-butanol yield stayed in the range of 10-14%. The slow decrease in activity after the first cycle was additionally accompanied by an evident reduction of the macroporosity seen from SEM images (Figure 6a and 6b). These changes were also confirmed by a change in the BET surface area from 255.7 in the fresh catalyst to 120.5 m²/g after reaction, and a reduction in pore volume from 1.06 to 0.67 cm³/g (Table 1, Entry 4).

Furthermore, as shown in the TEM images (Figure 6d), the formation of well dispersed nanoparticles with a diameter ranging between 20 – 50 nm could be observed, which can be attributed to the formation of active Cu⁰/Ni⁰ species²⁸. The main difference between the recovered catalyst after the first and the last recycling test (Figure 6d and 6e) was the partial agglomeration of smaller nanoparticles and the formation of several bigger nanoparticles with a diameter up to 100 nm. This gradual sintering might be eventually responsible for the reduced catalytic activity, although no major deactivation was apparent after 7 cycles. Additionally, ICP analysis (Table S7) showed almost no leaching of the Cu and Ni incorporated in the catalyst structure while a small amount of Mg and Al loss was observed after the first cycle and almost negligible amounts were detected in the subsequent cycles. This also confirms that the main modification of the catalyst structure takes place after the first cycle and the structure becomes robust during subsequent use.

Structure and reactivity

The excellent activity of the new Cu₁₀Ni₁₀-PMO catalyst can be attributed to several structural parameters. Both SEM (Figure 6a) and TEM measurements (Figure 6c) of the fresh catalysts showed porous structures that did not display any agglomeration of the copper and nickel dopants, according to a homogeneous dispersion of the transition metals into the catalyst structure. We expected, that the simultaneous presence of Cu and Ni may lead to the formation of a Cu-Ni alloy phase during the reaction which could favor the dehydrogenation of ethanol and enhance the stability of the catalyst due to the stronger interaction of both dehydrogenation metals.³⁵ Indeed, elemental mapping (Figure 6h) of a sample collected after the 4th run showed the formation of nanoparticles, which

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3 contain Cu and Ni. At least one particle in the area shown in Figure 6h indicates the formation of Cu-
4 Ni alloy.³⁶ This was additionally confirmed by the powder XRD pattern of the spent catalyst (Figure S1)
5 collected after the 1st and 4th cycle with the peak at 43.0° and 50.9° assigned to the formation of
6 crystalline Cu-Ni alloy regions, as previously also observed by Dalin³³ and Bonura³⁷.
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10 Based on previous reports^{13–15,17,18,24,38–41}, an additional parameter markedly influencing reactivity is
11 the acidity and basicity of these catalysts. The distributions of acidic and basic binding sites were
12 determined by temperature-programmed desorption (TPD) of NH₃ and CO₂ for selected different
13 PMO. The corresponding TPD profiles (Figure S2) and relative number of acid (Table S7) and basic
14 sites (Table S9) can be found in the supporting information. These measurements found that the Mg-
15 Al-PMO, not containing any dopants, showed the highest population of strong acidic sites (76%)
16 while the lowest was attributed to the Cu₁₀Ni₁₀-PMO composition (59%). Similarly, the MgAl-PMO
17 catalyst showed the highest basicity while the CO₂ uptake reached a minimum value (0.41 mmol/g_{cat})
18 for the Cu₁₀Ni₁₀-PMO catalyst. These more balanced acidity and basicity levels compared to the
19 parent MgAl-PMO hint towards a possible correlation between these parameters and reactivity and
20 influencing selectivity of 1-butanol. It is interesting that the rate of 1-butanol formation increased
21 linearly with the ratio of base/acid sites (Figure S3).
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24 *Coupling of higher n-alcohols*

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26 Finally, the self-coupling of several n-alcohols (n=3 to 6) was attempted in order to observe the
27 influence of the chain length on the extent of dimerization. The results are displayed in Figure S4. The
28 biggest difference between the reactivity of these alcohols compared to ethanol was the overall
29 lower conversion (below 30%) of the starting material. In the case of 1-propanol, a higher
30 ester/alcohol ratio was observed. In the self-coupling of 1-butanol, 1-pentanol and 1-hexanol, an
31 even lower substrate conversion was seen. Esters and ethers were obtained as major products and
32 only minor amount (<5% yield) of self-coupling product was found. This is in agreement with the
33 persistence of 1-butanol and higher alcohols in the reaction medium when ethanol is used as starting
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material. Nevertheless, the formation of butyric acid esters and butyl ethers is a major pathway for 1-butanol consumption.

Conclusion

In summary, we report herein copper- and nickel-doped porous metal oxide catalysts for the Guerbet coupling of ethanol to 1-butanol. A novel composition containing equimolar amounts of copper and nickel dopants afforded an especially active catalyst that led to 22% 1-butanol yield at 56% ethanol conversion at 320 °C for 6h, a result that is excellent among the known heterogeneous catalysts. The catalyst has proven robust without significant deactivation over 7 cycles. Structural characterization before and after reaction revealed the formation of metal nanoparticles as well as the existence of a Ni-Cu alloy phase that likely plays a role in the excellent catalytic performance.

Associated content

Supporting information

In the Supporting Information, additional details of the experimental procedures, characterization of catalysts, analysis of products and comparison of selected examples of heterogeneous catalysts systems are available.

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Table 1 Composition and textural properties of PMO catalysts prepared .

Entry	Catalyst code	Theoretical composition ^a	Experimental composition ^b	Surface area (m ² /g)	Pore volume (cm ³ /g)	Average pore size (nm)
1	MgAl-PMO	Mg _{3.0} Al _{1.0}	Mg _{3.0} Al _{1.0}	233.8	0.91	21.9
2	Ni20-PMO	Ni _{0.6} Mg _{2.4} Al _{1.0}	Ni _{0.6} Mg _{2.1} Al _{1.0}	239.2	0.74	17.6
3	Cu20-PMO	Cu _{0.6} Mg _{2.4} Al _{1.0}	Cu _{0.6} Mg _{2.4} Al _{1.0}	197.7	0.96	21.2
4	Cu10Ni10-PMO	Cu _{0.3} Ni _{0.3} Mg _{2.4} Al _{1.0}	Cu _{0.3} Ni _{0.3} Mg _{2.3} Al _{1.0}	255.7 (120.5) ^c	1.06 (0.67) ^c	17.3 (19.6) ^c

a. Based on quantities of metal ions used during co-precipitation. b. Determined by ICP analysis. c. Numbers in brackets refer to the catalyst recovered after reaction (Catalyst 200mg, ethanol 3ml, 310 °C, 6h).

Table 2 Screening of the various prepared PMO compositions in the Guerbet reaction of ethanol to 1-butanol.

Entry	Cat.	Conv. %	Yield%								STY (g _{pro} kg _{cat} ⁻¹ h ⁻¹) ^a	Alcohols /esters
			ButOH	AcOEt	HexOH	OctOH	EButOH	EB	EHexOH	AcOBut		
1	MgAl-PMO	30.8	5.6(83) ^b	0.3	0.6	0.1	0.1	0.1	0.0	0.0	176.3	16.0
2	Cu20-PMO	37.8	5.7(22)	18.1	0.0	0.0	0.0	0.9	0.0	1.0	180.4	0.3
3	Ni20-PMO	30.3	8.8(73)	0.5	1.2	0.1	0.3	0.1	0.0	0.0	278.0	17.3
4	Cu13Ni7-PMO	62.4	18.7(66)	1.5	4.0	0.8	1.0	0.8	0.3	0.5	594.3	8.9
5	Cu7Ni13-PMO	51.2	17.9(72)	0.7	3.8	0.8	0.9	0.6	0.3	0.0	567.8	18.2
6	Cu10Ni10-PMO	47.9	21.1(72)	0.9	4.5	0.7	0.9	0.7	0.4	0.3	670.3	13.8
7 ^c	Cu10Ni10-PMO	47.6	21.7(72)	1.6	4.3	0.6	1.1	0.7	0.2	0.0	670.3	12.1
8	Cu10Ni10-HTC	46.7	17.7(80)	0.8	2.5	0.1	0.8	0.2	0.2	0.0	561.7	21.2

Reaction conditions: Catalysts 0.1g, Ethanol 3ml, 310 °C, Decane 20μl as internal standard, 6h.

a. Space time yield of 1-butanol. b. Number in the bracket is the selectivity of 1-butanol. c. Reaction under Ar.

ButOH: 1-Butanol, AcOEt: Ethyl Acetate, HexOH: 1-Hexanol, OctOH: 1-Octanol, EButOH: 2-Ethylbutanol, EB:Ethyl Butyrate, EHexOH:2-Ethylhexanol, AcOBut: Butyl Acetate.

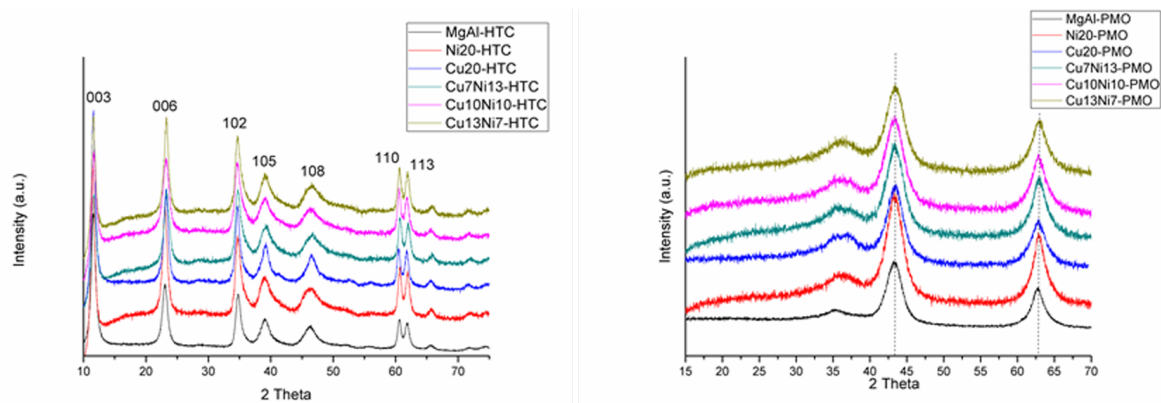


Figure 1 XRD patterns of the synthesized hydrotalcites (left) and the corresponding porous metal oxides (PMOs) after calcination (right).

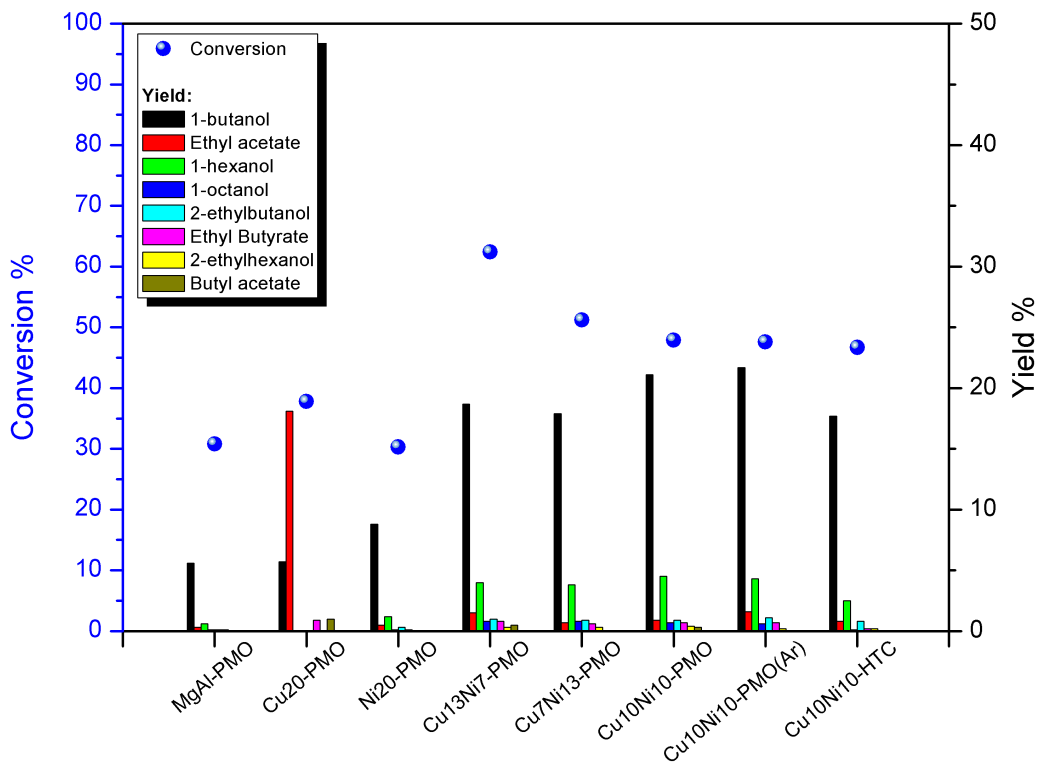


Figure 2 Screening of different PMO compositions in the Guerbet reaction of ethanol to 1-butanol. Reaction conditions: catalyst (100mg), ethanol (3ml), 6h, 310 °C, decane (20μl).

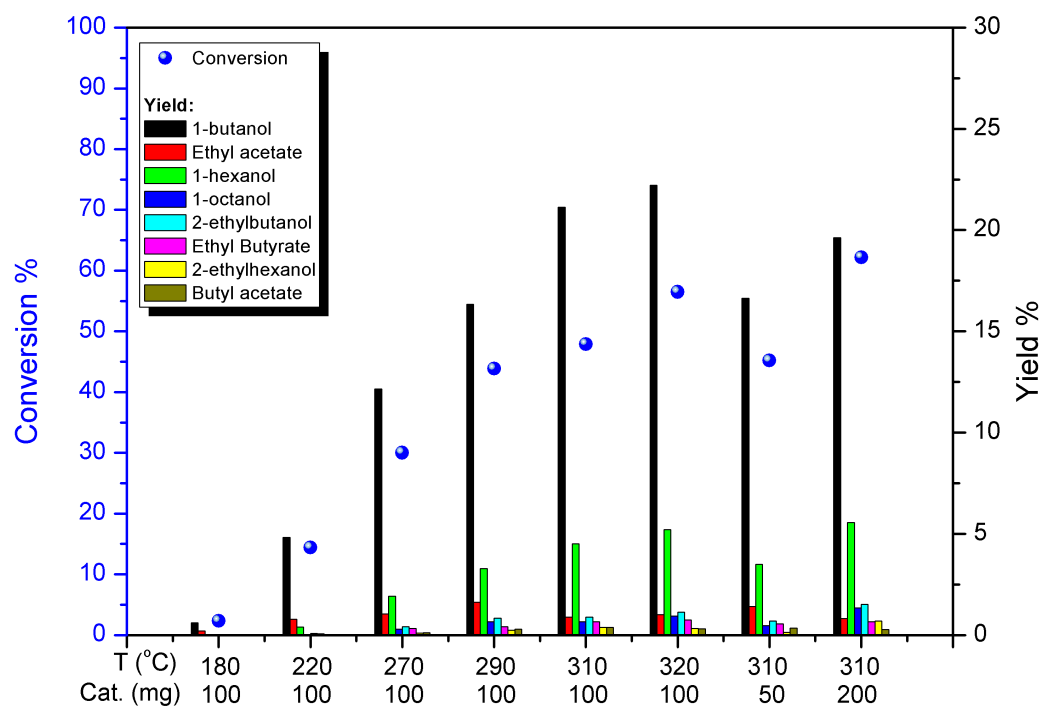


Figure 3 Influence of temperature and catalyst loading on ethanol conversion and distribution of products. Reaction conditions: Cu10Ni10-PMO (50-200mg), ethanol (3ml), 6h, decane (20μl).

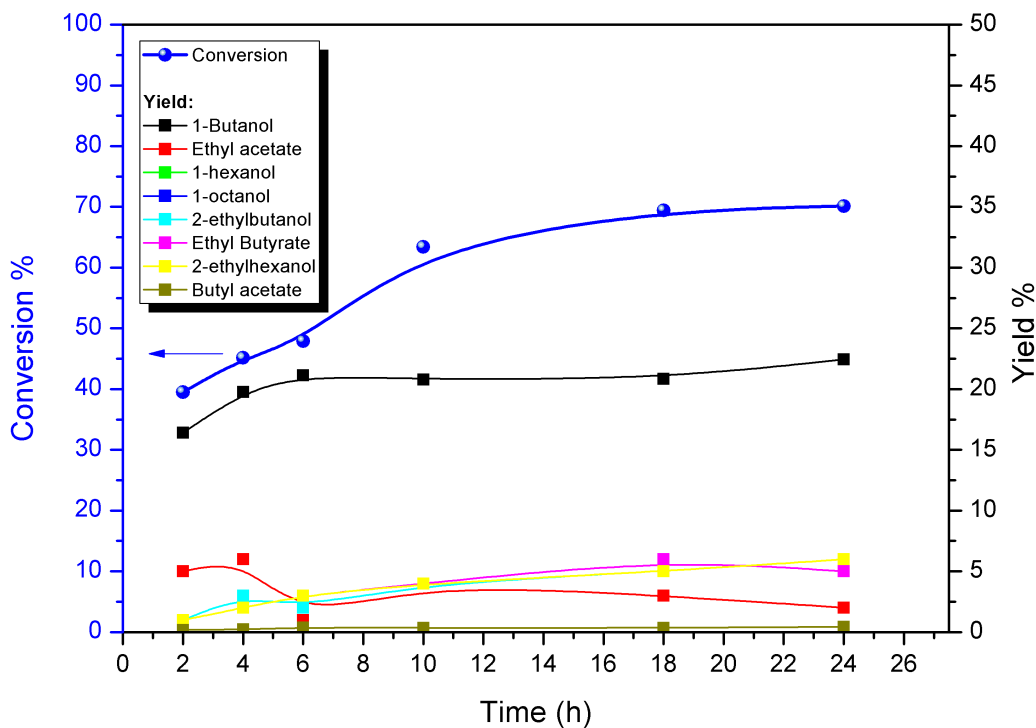


Figure 4 Product formation profile for the Guerbet reaction of ethanol to 1-butanol for 24 hours. Reaction conditions: Cu10Ni10-PMO (100mg), ethanol (3ml), 310 °C, decane (20μl).

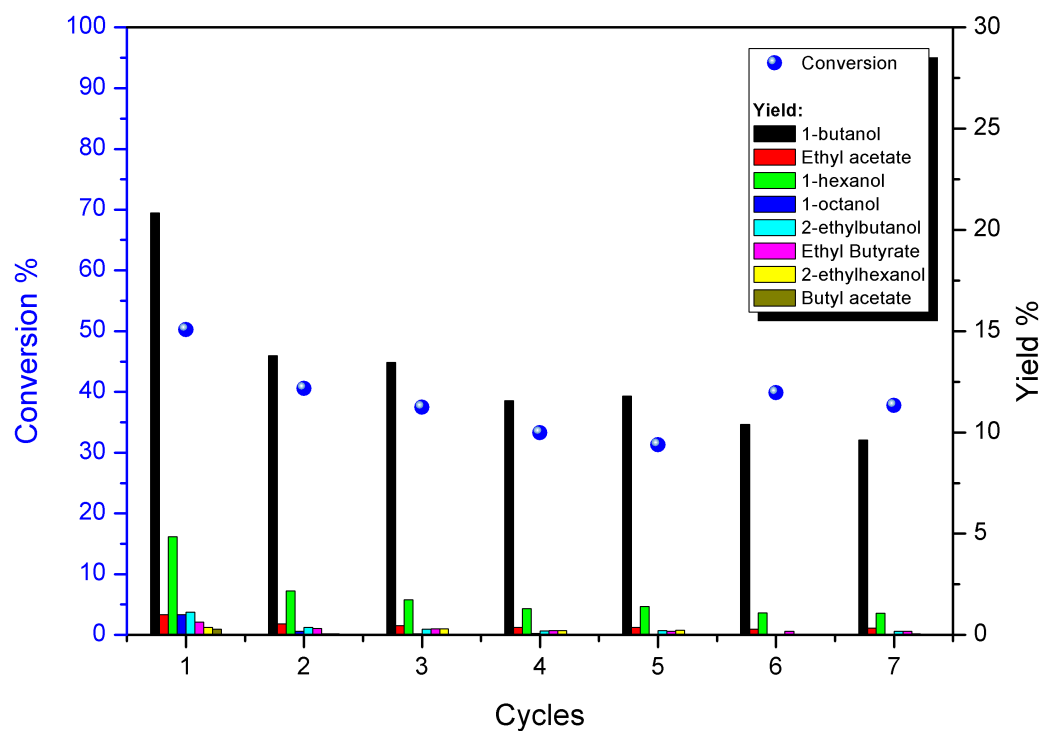


Figure 5 Recycling test. Reaction conditions: Cu10Ni10-PMO (100mg), ethanol (3ml), 310 °C, 6h, decane (20 μ l).

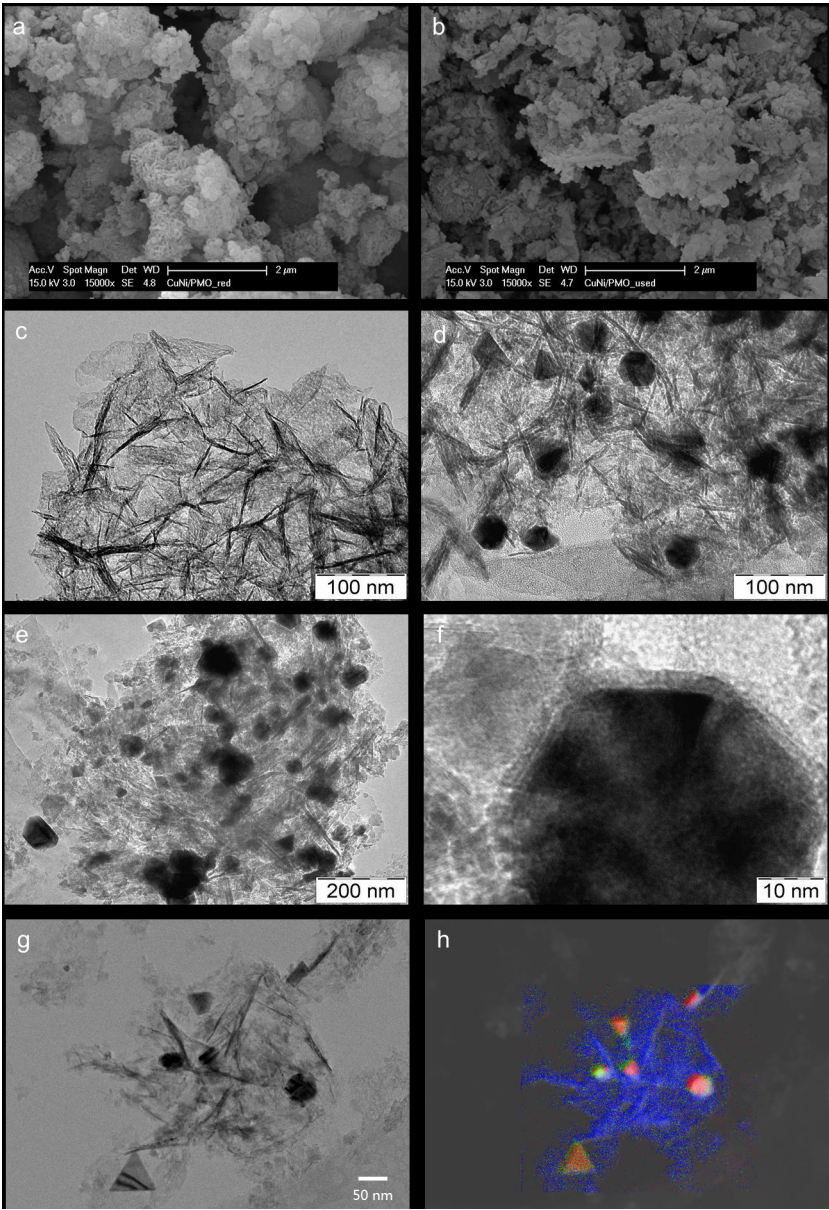
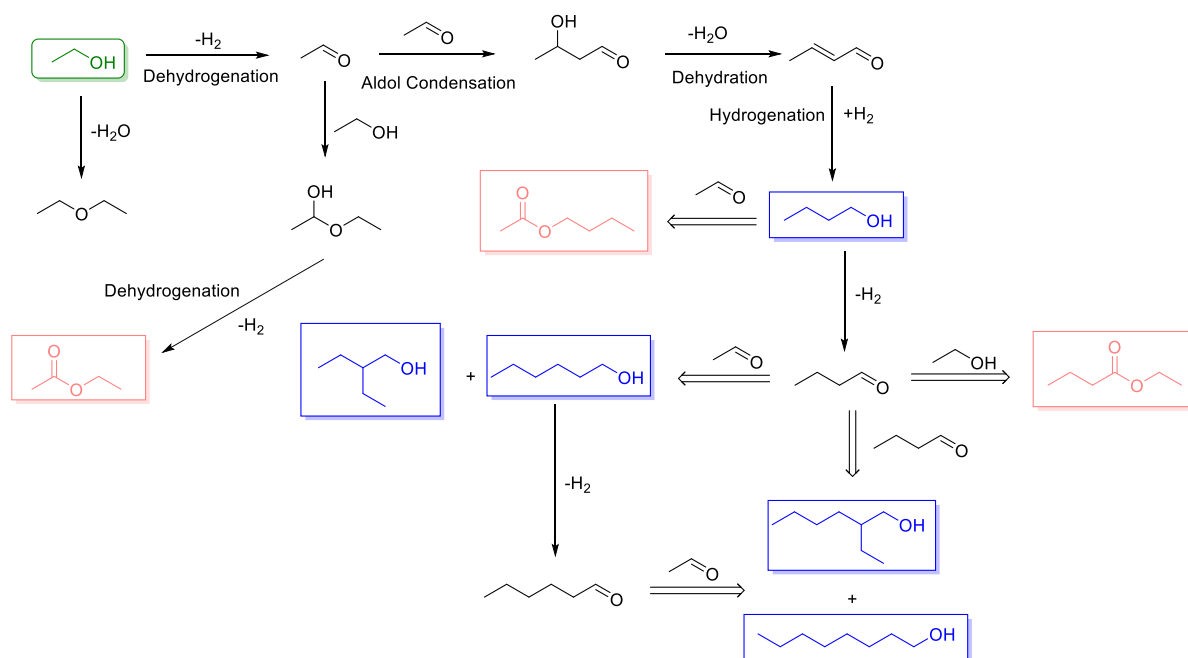
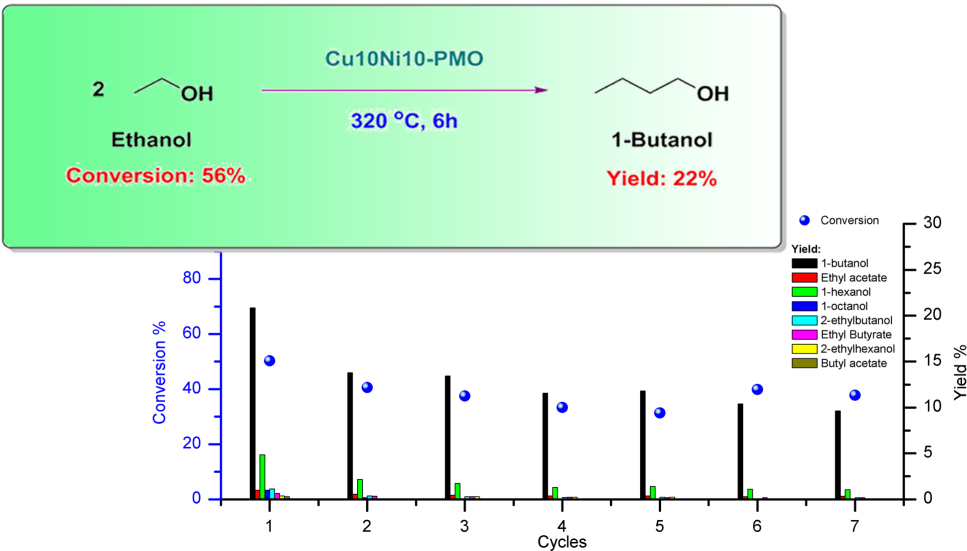


Figure 6 SEM images of Cu₁₀Ni₁₀-PMO catalyst before (a) and after one reaction (b) and TEM images of Cu₁₀Ni₁₀-PMO catalyst: (c) before reaction; (d) after one reaction; (e) and (f) after 7 cycles; (g) and (h) after 4 cycles. Color code refers to the distribution of elements (Cu: red, Ni: green, Mg: blue).



Scheme 1 Reaction network for Guerbet coupling of ethanol.

Table of contents



Title: Efficient catalytic conversion of ethanol to 1-butanol via the Guerbet reaction over copper- and nickel-doped porous metal oxides

Authors: Zhuohua Sun, Anaís Couto Vasconcelos, Giovanni Bottari, Marc C. A. Stuart, Giuseppe Bonura, Catia Cannilla, Francesco Frusteri, Katalin Barta

Synopsis: A novel porous metal oxide containing equimolar amounts of Cu and Ni dopants is reported as efficient catalyst for the Guerbet-coupling of ethanol to 1-butanol. The catalyst is selective and robust over several cycles.